Supplementary Material for Confidence-Shielded, Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

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1 Data provenance and split control

The official LLVIP partitions are larger than the experiment subsets. The 300-image test subset contains the first 300 pairs after a uniformly random shuffle (seed 0) of official test pairs with non-empty labels. The 1,500-image training subset comprises the lexicographically first 1,500 official training pairs with non-empty labels. Within this ordered subset, the first 1,350 pairs form training and the remaining 150 form validation. The manifest contains the complete stem lists and SHA-256 digests. All three pairwise stem intersections are empty. The test subset was not used for checkpoint selection or policy fitting.

Table 1: Partition sizes and annotation counts.

Dataset	Training	Validation	Test	Test groups
LLVIP	1,350	150	300	7 scenes
M3FD	1,600	400	200	2 frame groups
DroneVehicle	public train	public validation	300	3 frame groups

Using the same training-run-cluster and group bootstrap for the primary CS-ECF-versus-visible contrast gives a dataset-balanced AP50 gain of 0.0133 and a 95% interval of $[0.0026, 0.0261]$. This clustered interval is the primary uncertainty statement; treating the two DroneVehicle checkpoints as independent would understate their shared training-run dependence.

Table 2: LLVIP annotation counts in the fixed split and pairwise overlap.

Split	Pedestrian boxes	Pairwise stem overlap
Training	4,479	train–validation: 0
Validation	419	validation–test: 0
Test	744	train–test: 0

M3FD frames are partitioned into consecutive 100-frame groups. The initial grouping seed is 2028 and the extension seed is 2029; the two held-out test groups do not enter training or validation. DroneVehicle uses three consecutive 100-frame groups selected from the official test archive with seed 2037. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E`.

The YOLO11x-OBB best and last checkpoints share one 150-epoch training run (batch size 32, 640-pixel input, seed 0). They are treated as one training-run cluster in every aggregate bootstrap.

2 Detection evaluation

All predictions are generated at a confidence floor of 0.001 and a 640-pixel inference size. YOLO predictions use NMS at IoU 0.7. For each image, class and IoU threshold from 0.50 to 0.95 in

increments of 0.05, predictions are assigned to ground truth by one-to-one maximum-cardinality matching over all valid prediction–annotation pairs. The resulting true positives and confidence scores are pooled across the evaluated images, and AP is computed on the 101-point interpolated precision–recall curve with Ultralytics 8.4.189. AP50 averages class AP at IoU 0.50; AP50–95 additionally averages the ten IoU thresholds.

3 Complete CS-ECF and Always CDF results

The tables below reconstruct the frozen policy from released per-image detections. "Selected" gives the number of images assigned to visible, CDF, PIAFusion and SPGFusion. The group intervals are percentile bootstraps over scenes or consecutive frame groups.

Table 3: Per-checkpoint paired comparison on LLVIP. Delta is CS-ECF minus Always CDF; the last column is Visible/CDF/PIAFusion/SPGFusion selections.

Evaluator	CDF	CS-ECF	Delta	95% group CI	V/C/P/S
YOLO11n	.8920	.8845	−.0076	[−.024, +.012]	276/1/9/14
Agnostic	.8629	.8611	−.0018	[−.022, +.011]	286/2/6/6
YOLO26n s0	.8200	.8088	−.0112	[−.034, +.009]	288/1/3/8
RT-DETR-l	.8100	.8136	+.0036	[−.015, +.018]	293/0/0/7
YOLO26n s1	.8048	.7911	−.0138	[−.026, +.002]	265/1/4/30
YOLO26n s2	.8183	.8087	−.0096	[−.023, +.012]	282/0/0/18
YOLOv8n s0	.7847	.7761	−.0086	[−.025, +.011]	282/1/0/17
YOLOv8n s1	.8087	.8119	+.0032	[−.015, +.024]	290/0/0/10

Table 4: Per-checkpoint paired comparison on M3FD and DroneVehicle.

Dataset	Evaluator	CDF	CS-ECF	Delta	95% group CI
M3FD	YOLO26n	.3705	.3757	+.0053	[−.0002, +.0114]
M3FD	YOLOv8n	.3731	.3972	+.0241	[+.0087, +.0321]
DroneVehicle	best	.6624	.6814	+.0190	[−.0013, +.0301]
DroneVehicle	last	.6524	.6767	+.0243	[−.0067, +.0464]

Table 5: Paired CS-ECF minus Always CDF summary.

Scope	Difference	95% interval
LLVIP	−0.0057	[−0.0189, +0.0156]
M3FD	+0.0147	[+0.0100, +0.0159]
DroneVehicle	+0.0217	[−0.0007, +0.0738]
Dataset-balanced	+0.0102	[−0.00057, +0.02980]

The combined interval resamples scene or frame groups and training-run clusters within each dataset, then assigns equal weight to the three datasets. There are eight run clusters on LLVIP, two on M3FD and one on DroneVehicle. The DroneVehicle component therefore quantifies test-group uncertainty but cannot estimate variability between independent DroneVehicle training runs. The lower endpoint of the combined interval is below zero, so the direct CDF comparison does not establish a separated average improvement.

3.1 Class-specific M3FD results

The 200-image M3FD test split contains 2,022 annotations. AP50 is the mean of the six class AP values, so class frequency does not determine a class's weight in the headline metric.

Table 6: M3FD class-specific CS-ECF minus visible AP50. Intervals are 95% percentile image-bootstrap intervals from 400 paired resamples. The test split has only two frame groups, so these class intervals are descriptive and are not presented as stable grouped inference.

Class	Annotations	YOLO26n	95% interval	YOLOv8n
People	166	+0.0214	[−0.0070, +0.0498]	+0.1090
Car	1,597	0.0000	[0.0000, 0.0000]	−0.0001
Bus	16	0.0000	[0.0000, 0.0000]	0.0000
Motorcycle	189	0.0000	[0.0000, 0.0000]	0.0000
Lamp	20	+0.0030	[0.0000, +0.0102]	+0.0001
Truck	34	0.0000	[0.0000, 0.0000]	+0.0003

For YOLOv8n, the people-class image interval is [+0.0440, +0.1913]. The people class accounts for 87.7% of the positive class-mean contribution for YOLO26n and 99.6% for YOLOv8n. Because the class-balanced metric gives people one-sixth of the total weight despite its small annotation count, this concentration directly affects the six-class AP50.

4 External evaluator stress tests

Two COCO-pretrained detectors, YOLOv9t and YOLOv10s, were used without LLVIP fine-tuning. They were not used in policy construction, threshold selection, or the primary evaluation. Person-class detections were retained before AP and selection scoring. The 1.5 margin and top-five score were unchanged.

A third stress test used an LLVIP-fine-tuned YOLOv9t. The checkpoint was initialized from COCO YOLOv9t and trained on the 1,350 visible training images for a pre-specified 40 epochs. Training used AdamW, initial learning rate 0.002, three-epoch warm-up, cosine decay, weight decay 5×10^{-4}, batch size 16, 640-pixel inputs, random horizontal flipping, and brightness/contrast jitter. A reproducible PIL-based loader calling the Ultralytics detection loss was used. The final checkpoint after epoch 40 was frozen before the LLVIP test images were scored; no epoch was selected using test results. Its SHA-256 is `AE0A44359C489DEEAE3F7443173588B9F68E264B6BE1EA455E961EB2DED0797D`. The checkpoint reached mAP50 0.8456 on the 150-image validation set and 0.6666 on the 300-image test visible input.

Table 7: External evaluator stress tests on the same 300 LLVIP test pairs.

Model	Visible	Always CDF	No-shield	CS-ECF
YOLOv9t, zero-shot	.7698	.7888	.8216	.7948
YOLOv10s, zero-shot	.8344	.8617	.8699	.8557
YOLOv9t, fine-tuned	.6666	.6598	.6405	.6558

Table 8: Paired confidence intervals and scene-tail behaviour for the external evaluators. The final column reports the worst scene change for CS-ECF / no-shield.

Model	CS-ECF − visible	CS-ECF − CDF	Worst (CS/NS)
YOLOv9t, zero-shot	+.0250 [+.0053, +.0523]	+.0060 [−.0277, +.0374]	−.0167/ − .0011
YOLOv10s, zero-shot	+.0213 [+.0041, +.0439]	−.0059 [−.0266, +.0180]	−.0038/ − .0337
YOLOv9t, fine-tuned	−.0108 [−.0342, +.0050]	−.0040 [−.0206, +.0108]	−.0724/ − .1208

The shield selected visible on 233/300 zero-shot YOLOv9t, 245/300 YOLOv10s, and 230/300 LLVIP-fine-tuned YOLOv9t images. The two zero-shot evaluators support a positive difference from visible but not superiority over Always CDF. The fine-tuned YOLOv9t is a negative portability case: CS-ECF was below both visible and Always CDF, with scene-group intervals crossing zero. The shield improved the no-shield selector by 0.0153 AP50 (group interval [+0.0072, +0.0267]) and

reduced the worst scene loss from 0.1208 to 0.0724, but did not remove the loss. Tail behaviour therefore remains evaluator dependent rather than universally controlled.

Table 9: External-evaluator calibration and switching diagnostics over 300 LLVIP pairs. ECE is detection-level expected calibration error with ten fixed confidence bins. I/U/D denotes improved/unchanged/degraded per-image AP50 among images assigned to a fused candidate.

Model	Visible/fused	ECE V/CS	Switched I/U/D	Scenes	Score ratio
YOLOv9t, zero-shot	77.7%/22.3%	.009/.009	46.3/32.8/20.9%	5/2	2.28
YOLOv10s, zero-shot	81.7%/18.3%	.010/.012	38.2/52.7/9.1%	4/3	2.04
YOLOv9t, fine-tuned	76.7%/23.3%	.048/.046	25.7/32.9/41.4%	2/5	2.16

The score ratio is the median top-five-score ratio to visible among switched images. All three models satisfy the 1.5 selection margin at similar switch rates, but the fine-tuned checkpoint has a larger calibration error and a much higher fraction of harmful switches. Across the three external contrasts, the Benjamini–Hochberg q values are 0.043, 0.043, and 0.119 for YOLOv9t zero-shot, YOLOv10s zero-shot, and YOLOv9t LLVIP-fine-tuned, respectively. The corresponding detection-level Brier scores for visible and CS-ECF were 0.015/0.013, 0.013/0.011, and 0.087/0.086, respectively.

5 Validation choice, multiplicity, and exploratory alternatives

The policy was fixed before the confirmatory comparisons. The candidate set was retained to preserve the visible, region-selected, learned, and published fusion families. The top-five score was fixed before margin selection. On the 150 LLVIP and 400 M3FD validation images, the four evaluators use held-out groups and no confirmatory-test labels.

Table 10: Complete pre-freeze margin search for the four-candidate top-five selector on validation data. Balanced gain and worst evaluator gain are over the four validation evaluators. Positive counts are evaluator-level point estimates. Margin 1.5 was the first setting satisfying the 0.0005 per-evaluator loss bound and had the largest balanced gain among feasible settings.

Margin	Balanced gain	Worst gain	Positive
1.00	-0.0064	-0.0249	1/4
1.10	-0.0019	-0.0104	1/4
1.20	-0.0002	-0.0056	2/4
1.30	+0.0000	-0.0026	2/4
1.50	**+0.0008**	**-0.0001**	**3/4**
1.75	-0.0006	-0.0015	1/4
2.00	+0.0001	-0.0004	2/4

For the frozen $K = 5$ rule, margin 1.5 was the first setting meeting the 0.0005 per-evaluator loss bound and had the largest balanced gain in the complete pre-freeze grid. Its validation gains were 0.001850 and -0.000070 for the two LLVIP evaluators and 0.001518 and 0.000083 for the two M3FD evaluators. The post-freeze test grid contains settings with higher observed means, but those settings were evaluated after the primary policy was frozen and are not used to redefine it.

The test grid is not a model-selection procedure. Some unselected settings have larger means, but several also increase switching or tail exposure. In particular, $K = 10$, margin 1.5 has a larger test mean but only 7/12 positive checkpoints and a worst loss of 0.0119; $K = 10$, margin 2.0 has a similar tail to CS-ECF and a larger mean in this post-hoc check. These observations motivate prospective validation of alternative score depths and do not retroactively change the frozen primary policy.

5.1 Multiplicity analysis

The primary inference family contains the 12 prespecified CS-ECF-versus-visible checkpoint contrasts. Two-sided p-values come from 2,000 pairwise scene/frame-group bootstrap resamples per checkpoint. No checkpoint had a Holm-adjusted p-value below 0.05. The best raw result was LLVIP/YOLO11n, with a gain of

Table 11: Post-freeze test sensitivity over score depth K and visible margin for the full four-candidate library. Values are dataset-balanced AP50 gain, worst checkpoint mean gain, positive checkpoints, and mean switched-decision fraction. These results are descriptive and were not used to select the policy.

K	Margin	Balanced gain	Worst	Positive	Switch (%)
1	1.2	+0.0101	-0.0012	10/12	7.89
1	1.5	+0.0088	-0.0004	11/12	5.32
1	2.0	+0.0095	-0.0007	11/12	3.79
5	1.2	+0.0182	-0.0276	7/12	18.25
5	1.5	**+0.0133**	**-0.0040**	9/12	8.63
5	2.0	+0.0099	-0.0010	10/12	5.19
10	1.2	+0.0268	-0.0377	7/12	26.19
10	1.5	+0.0218	-0.0119	7/12	12.94
10	2.0	+0.0169	-0.0019	9/12	7.11

+0.0034, raw $p = 0.0050$, and Holm-adjusted $p = 0.0600$. The exact sign test for 9/12 positive checkpoints gave $p = 0.0730$. These results do not support a claim that a specific checkpoint improved significantly; they support only the prespecified balanced estimand and the observed risk reduction.

Table 12: Checkpoint-level visible-relative contrasts and multiplicity. Intervals resample scene or consecutive frame groups. Raw p-values are two-sided bootstrap probabilities; Holm and BH values are adjusted for the 12-member confirmatory family.

Checkpoint	Gain	95% group interval	Raw p	Holm	BH
LLVIP/YOLO11n	+0.0034	[+0.0011,+0.0056]	0.0050	0.0600	0.0600
LLVIP/Agnostic	+0.0061	[+0.0002,+0.0193]	0.0310	0.3408	0.1859
LLVIP/YOLO26n s0	-0.0034	[-0.0092,+0.0017]	0.4038	1.0000	0.6433
LLVIP/RT-DETR-l	+0.0040	[+0.0000,+0.0073]	0.1889	1.0000	0.3778
LLVIP/YOLO26n s1	-0.0027	[-0.0139,+0.0091]	0.6847	1.0000	0.6847
LLVIP/YOLO26n s2	+0.0026	[-0.0027,+0.0142]	0.5877	1.0000	0.6433
LLVIP/YOLOv8n s0	-0.0040	[-0.0164,+0.0032]	0.5897	1.0000	0.6433
LLVIP/YOLOv8n s1	+0.0064	[+0.0000,+0.0202]	0.0510	0.5097	0.1979
M3FD/YOLO26n	+0.0041	[+0.0000,+0.0058]	0.5177	1.0000	0.6433
M3FD/YOLOv8n	+0.0182	[+0.0000,+0.0256]	0.4908	1.0000	0.6433
DroneVehicle/best	+0.0251	[-0.0003,+0.0387]	0.0660	0.5937	0.1979
DroneVehicle/last	+0.0293	[-0.0014,+0.0479]	0.1589	1.0000	0.3778

Two alternative policies were evaluated after the frozen comparison. They are exploratory and are not used to support the main claim. A top-10 selector over visible/CDF/PIAFusion with margin 1.3 gave a 0.0039 balanced gain but a hierarchical interval of $[-0.0029, +0.0127]$. A mean-confidence selector over visible/SPGFusion with margin 1.5 gave a -0.0034 balanced gain and $[-0.0085, +0.0013]$. Both alternatives fail to establish a stable advantage across the 10 LLVIP/M3FD checkpoints.

6 Registration sensitivity

ECC translation residuals were computed on normalized Sobel edges for all 300 LLVIP test pairs. The median residual was 2.53 pixels. The best-aligned and worst-aligned quartiles each contain 75 pairs; the best-quartile residual cutoff is 1.61 pixels.

Table 13: CS-ECF-minus-visible AP50 over LLVIP registration quartiles, summarised over eight checkpoints. Each checkpoint interval uses 2,000 image-bootstrap resamples within the selected quartile.

Quartile	Mean gain	Worst	Best	Positive
Best aligned	+0.0007	-0.0125	+0.0128	5/8
Worst aligned	+0.0020	-0.0237	+0.0150	6/8

The gain is not confined to the best-aligned subset. Most checkpoint-specific intervals still cross zero, however, and the result should not be read as a significant registration effect. Equivalent residual estimates were not available for M3FD or DroneVehicle.

7 Thermal controls

Table 14: Spearman association between switching and thermal or visible features. These secondary association tests are descriptive and are not multiplicity-adjusted.

Dataset	Thermal top-hat ρ (p)	Visible luma ρ (p)	CDF-mask ρ (p)
LLVIP	$+0.151$ (0.009)	-0.104 (0.071)	-0.029 (0.614)
M3FD	$+0.320$ ($< 10^{-5}$)	-0.066 (0.356)	-0.332 ($< 10^{-5}$)
DroneVehicle	-0.403 ($< 10^{-5}$)	-0.441 ($< 10^{-5}$)	$+0.064$ (0.269)

In the fixed-input CDF ablation on DroneVehicle, the thermal mask changed AP50 from 0.6620 to 0.6624 for the best checkpoint and from 0.6512 to 0.6524 for the last checkpoint, relative to global detail injection.

The association reverses between datasets. Thermal contrast is therefore not treated as a sufficient proxy for fusion utility, and no universal thermal-switching rule is claimed.

8 Reproduction

The public repository contains the selector, candidate-generation and detector scoring code, per-image score arrays, split manifests, source data for every figure and the scripts that reproduce the intervals. The direct baseline comparison is generated by:

```
python scripts/analyze_csecf_vs_cdf.py
```

This command validates each checkpoint AP before writing `results/selective_fusion/csecf_vs_cdf_paired.json`. Split and candidate manifests retain their seeds and selected groups. The multiplicity and sensitivity analysis is reproduced by:

```
python analysis/cs_ecf/major_revision_statistics.py \
  --data-root "$SSGF_DATA_ROOT" --out results/revision_statistics.json
```

Image-statistics controls, external-evaluator diagnostics, and registration quartiles use `analysis/cs_ecf/image_quality_baselines.py`, `analysis/cs_ecf/external_evaluator_diagnostics.py`, and `analysis/cs_ecf/alignment_csecf_sensitivity.py`, respectively. The image quality script requires `--external-root` for the image directories. M3FD class counts and contribution shares are generated by `analysis/cs_ecf/summarize_m3fd_classes.py`.